

Algebra 1: Polynomials

Foundation Level

Name: _____

Date: _____

Question 1: Solve for x in the equation $x^2 + 5x + 6 = 0$

A) $x = -2, x = -3$

B) $x = -1, x = -6$

C) $x = 2, x = 3$

D) $x = 1, x = 6$

E) $x = -2, x = 3$

Question 2: Factor the polynomial $x^2 - 7x + 12$

A) $(x - 3)(x - 4)$

B) $(x + 3)(x + 4)$

C) $(x - 2)(x - 6)$

D) $(x + 2)(x + 6)$

E) $(x - 1)(x - 12)$

Question 3: Simplify the expression $(2x^2 + 3x - 1) + (x^2 - 2x - 3)$

A) $3x^2 + x - 4$

B) $3x^2 + x - 2$

C) $3x^2 - x - 4$

D) $x^2 + x - 4$

E) $x^2 - x - 2$

Question 4: Find the x -intercept of the polynomial $2x^2 - 5x - 3$

A) $x = -$

rac12, $x = 3$

A) $x =$

rac12, $x = -3$

A) $x = -$

rac32, $x = 1$

A) $x =$

rac32, $x = -1$

A) $x = -1, x =$

rac32

Question 5: Determine the degree of the polynomial $3x^4 - 2x^2 + 1$

- A) 2
- B) 3
- C) 4
- D) 5
- E) 6

Question 6: Factor out the greatest common factor (GCF) from $12x^2 + 18x$

- A) $6x(2x + 3)$
- B) $6(2x^2 + 3x)$
- C) $12(x^2 +$

$\frac{3}{2}x)$

- C) $2(6x^2 + 9x)$
- D) $3(4x^2 + 6x)$

Question 7: Simplify the expression $(x^2 - 4)/(x + 2)$

- A) $x - 2$
- B) $x + 2$
- C) $x^2 + 2$
- D) $x^2 - 2$
- E) $x - 4$

Question 8: Find the value of x for which $x^2 + 4x + 4 = 0$

- A) $x = -1$
- B) $x = -2$
- C) $x = 1$
- D) $x = 2$
- E) $x = 0$

Question 9: Solve the quadratic equation $x^2 + 2x - 6 = 0$ using factoring, and show all your work and reasoning.

Question 10: Graph the quadratic function $f(x) = x^2 - 4x - 3$ and identify its x -intercepts, y -intercept, and vertex, showing all your work and reasoning.

Chef's Tips (Hints):

- When solving quadratic equations, always check your solutions by plugging them back into the original equation.
- To factor a quadratic expression, look for two numbers whose product is the constant term and whose sum is the coefficient of the linear term.
- When graphing a quadratic function, the vertex form $f(x) = a(x - h)^2 + k$ can be helpful in identifying the vertex and axis of symmetry.